

Interpreting the Latent Space of an ECG Digital-Twin Foundation Model

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A digital twin adapts an ECG foundation model and reveals its latent encodes infarct biophysics — but synthetic and real ECGs are intrinsically non-aligned, so the concept basis does not transfer to real ECGs

I • Classification

A shared head lifts synthetic→real transfer
0.56 → **0.76** AUC

II • Alignment

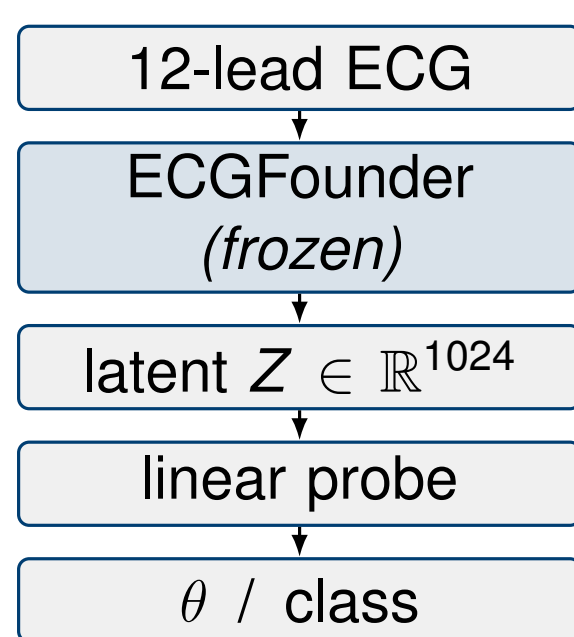
The synth→real gap is *intrinsic*: **C2ST** = 1.0, 3 methods, every K

III • Interpretability

Latent encodes infarct biophysics (**AUC 0.92**) — but it doesn't transfer

Background & method

A frozen foundation model (ECG-Founder) is adapted with **synthetic digital-twin** ECGs (MedalCare-XL, known ground-truth physics θ) and tested on **real** ECGs (PTB-XL).



Probes are *linear*; the backbone is frozen — anything decodable was already in the pretrained representation.

Conclusion

The twin **adapts** the model and acts as a **ground-truth oracle**: the latent encodes infarct biophysics in-domain. **But** synthetic and real ECGs are intrinsically non-aligned, so the concept *basis* does not transfer — the twin interprets the model, it is not yet a bridge to real ECGs.

Research plan

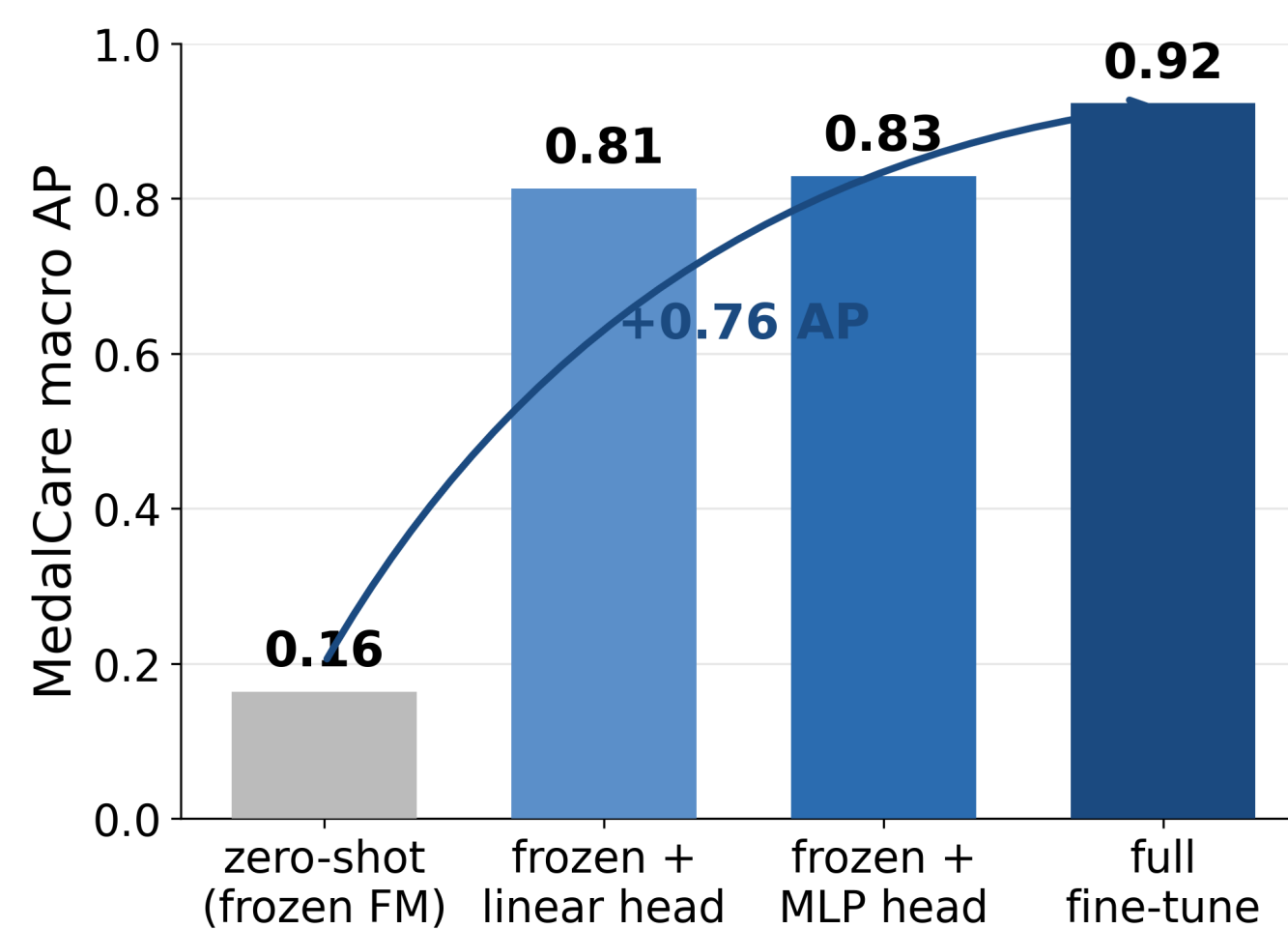
- **Multi-seed** ($n \geq 5$) to harden every claim.
- **Recover transfer**: multi-task bottleneck ($K=16$, loss-mix, curriculum).
- **Twin retrieval**: recover infarct location + action-potential parameters in one forward pass, validated on PTB-XL.
- **Counterfactual ECGs** (stretch): how an ECG changes as physiology varies.

A better latent on demand

A tiny $K=64$ bottleneck *trained to keep physics* **doubles** biophysical decoding and beats the full 1024-d latent — at no classification cost.

Decoded	1024-d	$K=64$
φ angle (R^2)	0.47	0.66
transmurality (AUC)	0.92	0.98

1 • Out of the box the model fails; adapting it recovers the signal



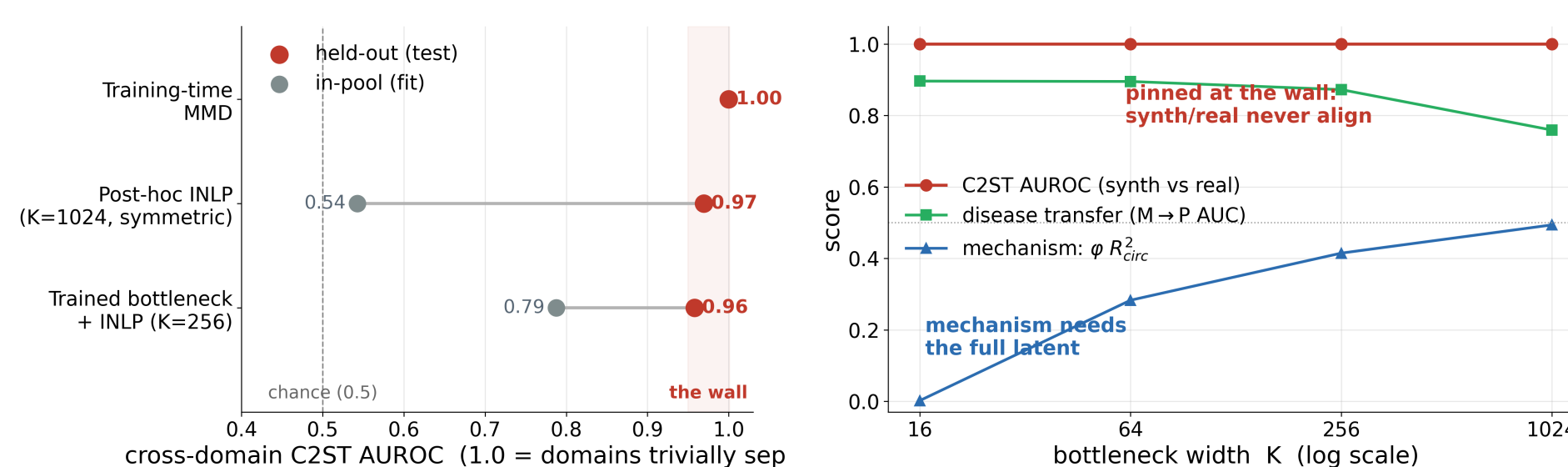
One linear head turns a near-useless zero-shot model into a strong one.

2 • A shared head drives synthetic→real transfer

Configuration	PTB-XL F1	M→P AUC
Joint dual-head	0.71	0.56
+ adapters	0.71	0.56
+ MMD alignment	0.70	0.59
Shared 3-class head	0.79	0.76

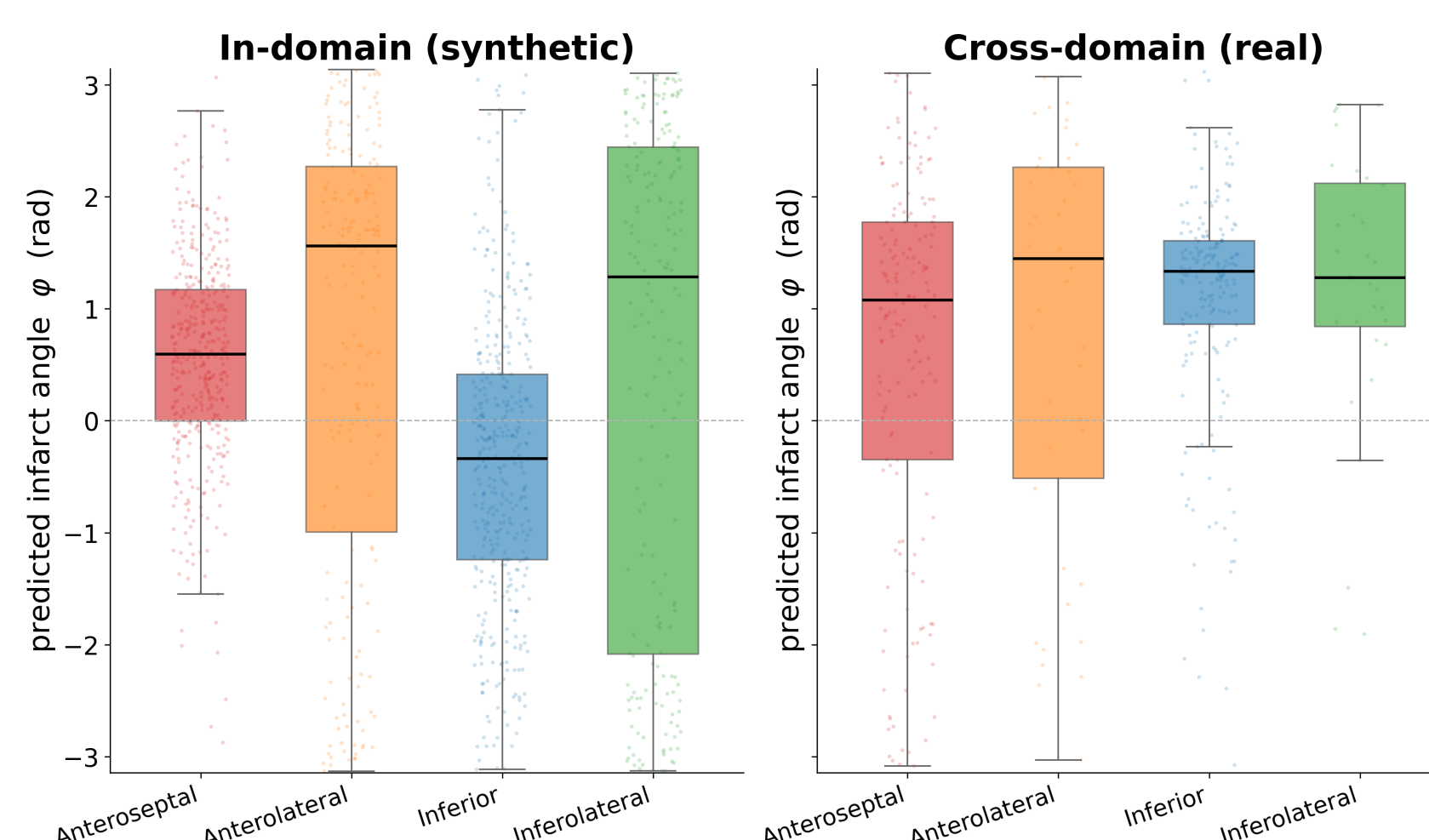
A shared classification head — not relabeling — nearly doubles synthetic→real transfer.

3 • The synthetic→real gap is intrinsic, not removable



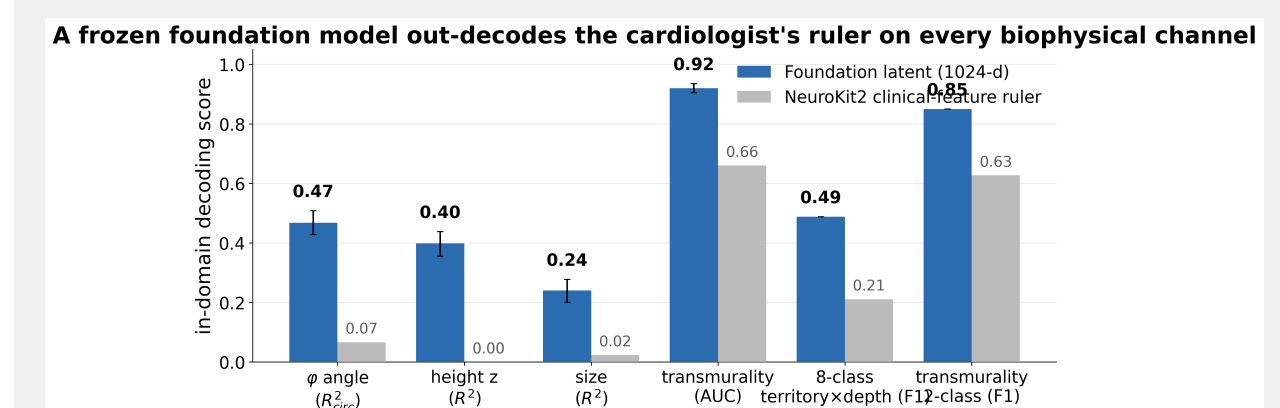
Three independent attacks all hit the same **C2ST** = 1.0 wall (left); shrinking the latent destroys mechanism but never closes the gap (right).

4 • The latent encodes biophysics — but the basis breaks on real ECGs



Infarct-angle decoding is sharp on synthetic ECGs but **collapses** on real ones.

Evidence: the latent encodes physics



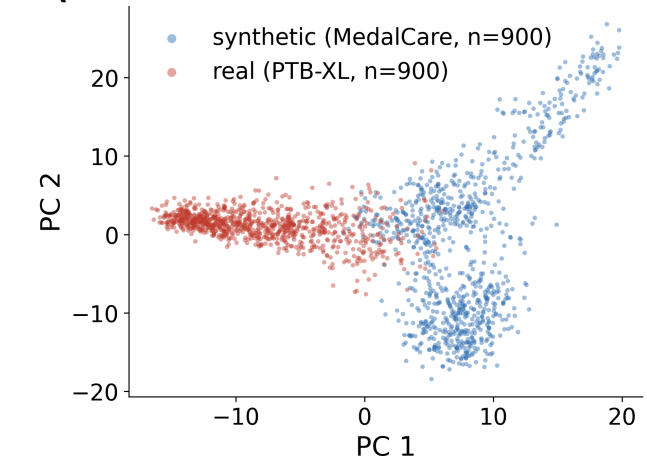
Frozen latents beat a clinical-feature ruler on every biophysical channel.

In-domain	Lat.	Rul.	ch.
8-cl. anat×depth	0.49	0.21	0.13
4-cl. anatomy	0.51	0.30	0.25
Transmurality	0.85	0.63	0.50

It even reads infarct *depth* — a ruler cannot.

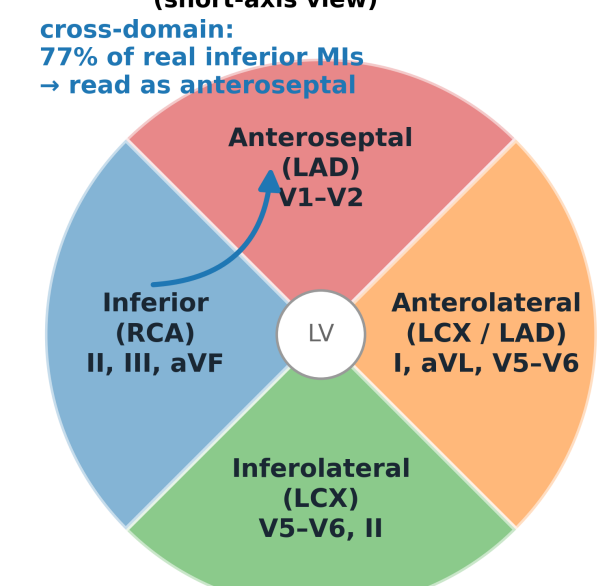
Why it fails: domains & anatomy

Synthetic and real ECGs occupy separate latent regions (domain classifier C2ST AUROC ≈ 1.0)



Synthetic and real latents form separate clouds.

Coronary territories on the left ventricle (short-axis view)



On real ECGs, **77%** of inferior infarcts are mistaken for anteroseptal ones.

True territory (PTB-XL)	Anteroseptal (n=168)	Anterolateral (n=42)	Inferior (n=196)	Inferolateral (n=32)
	51% (85)	18% (31)	17% (29)	14% (23)
	48% (20)	12% (5)	24% (10)	17% (7)
	77% (151)	4% (8)	17% (33)	2% (4)
Predicted territory	Anteroseptal	Anterolateral	Inferior	Inferolateral
	72% (23)	0% (0)	16% (5)	12% (4)

Across all four territories, cross-domain predictions are no better than chance.